Derivation of Pushing-Medium Calibration Constants: μ_G, k_{TT}, and k_{Fizeau}

Pushing-Medium Project

March 2026

Abstract

Three numerical constants appear throughout the Pushing-Medium (PM) numerical test suite and comparison code: the gravitational index coefficient $\mu_G = 1.4861 \times 10^{-27}\,\mathrm{m}^2\,\mathrm{kg}^{-1}$, the transverse-traceless power normalisation $k_{\mathrm{TT}} = 1.0$, and the Fizeau drag coupling $k_{\mathrm{Fizeau}} = 1.0$. This document records exactly how each value is obtained, the underlying physical reasoning, and how each value connects to known General Relativity (GR) results. The numbers are reproducible from first principles via the calibration fixture in `legacy/Pushing-Medium/tests/fixtures/calibration.py`.

1 Background: The PM Index Field for Gravity

In the Pushing-Medium framework, spacetime is modelled as a medium whose refractive index n encodes the gravitational field. For a point mass M the index at distance r is

$$n(r) = 1 + \frac{\mu_G\, M}{r}, \tag{1}$$

where μ_G is the gravitational index coefficient (a coupling constant with SI units $\mathrm{m}^2\,\mathrm{kg}^{-1}$). Light rays are governed by Fermat's principle in this medium: the deflection angle accumulated along a straight reference path at impact parameter b is

$$\alpha_{\mathrm{num}}(\mu_G) = \int_{-z_{\max}}^{+z_{\max}} \frac{\partial \ln n}{\partial x}\bigg|_{x=b,\, y=0} dz = \int_{-z_{\max}}^{+z_{\max}} \frac{-\mu_G M\, b}{n(r)\, r^3}\, dz, \tag{2}$$

where $r = \sqrt{b^2 + z^2}$. The analytic weak-field GR (Schwarzschild) result that this must reproduce is

$$\alpha_{\mathrm{GR}} = \frac{4GM}{c^2 b}. \tag{3}$$

2 Calibration of μ_G

2.1 Theoretical expectation

Comparing the leading-order expansion of (2) (with $n \approx 1$ in the denominator) to the GR result (3) immediately gives

$$\alpha_{\text{num}} \approx \mu_G M \int_{-\infty}^{+\infty} \frac{b}{(b^2 + z^2)^{3/2}}\, dz = \mu_G M \cdot \frac{2}{b}. \tag{4}$$

Setting this equal to (3) yields

$$\mu_G^{\text{theory}} = \frac{2G}{c^2} = \frac{2 \times 6.67430 \times 10^{-11}}{(2.99792458 \times 10^8)^2} \approx 1.4862 \times 10^{-27}\ \text{m}^2\,\text{kg}^{-1}. \tag{5}$$

2.2 Numerical determination procedure

Because the full integrand in (2) retains the factor $1/n(r)$ rather than approximating $n \approx 1$, the exact numerical value of μ_G is obtained by a bisection search rather than the closed-form expression (5). The procedure is:

1. Fix a solar test case: lens mass $M_\odot = 1.989 \times 10^{30}$ kg, impact parameter $b_\odot = 6.96 \times 10^8$ m (solar radius).
2. Compute the GR target: $\alpha^* = 4GM_\odot/(c^2 b_\odot) \approx 8.481 \times 10^{-6}$ rad ($\approx 1.75''$).
3. Bracket μ_G in the interval $[0.25 \times 2G/c^2,\ 4.0 \times 2G/c^2]$.
4. Run 36 bisection iterations, each evaluating $\alpha_{\text{num}}(\mu_G)$ numerically via `index_deflection_numeric` with $z_{\max} = 200\, b_\odot$, $N = 2500$ steps.
5. Return the midpoint of the final bracket.

The result converges to

$$\mu_G = 1.4861356300677034 \times 10^{-27}\ \text{m}^2\,\text{kg}^{-1}. \tag{6}$$

The relative difference from the analytic estimate (5) is $< 0.01\%$; the small discrepancy arises from the $1/n$ correction in the integrand and the finite integration window.

2.3 Physical interpretation

$\mu_G = 2G/c^2$ is the Schwarzschild radius per unit mass divided by c. Equivalently, it is the coefficient that makes the PM index gradient reproduce the standard PPN $\gamma = 1$ gravitational lensing result. Its appearance at exactly $2G/c^2$ is not accidental: it is the value required so that Fermat's principle in the medium (1) is equivalent, to leading order, to null geodesics in the linearised Schwarzschild metric.

3 Calibration of k_{TT}

3.1 Role of k_{TT}

The PM theory predicts gravitational-wave energy emission through a transverse-traceless (TT) quadrupole formula. The PM expression for the radiated power from a circular binary (masses M_1, M_2, separation a) is

$$P_{\mathrm{PM}} = \frac{32}{5}\,\frac{G^4\,(M_1 M_2)^2 (M_1 + M_2)}{c^5\,a^5}. \tag{7}$$

The GR (Peters 1964) formula for the same system is

$$P_{\mathrm{GR}} = \frac{32}{5}\,\frac{G^4\,(M_1 M_2)^2 (M_1 + M_2)}{c^5\,a^5}. \tag{8}$$

The normalisation factor is defined as

$$k_{\mathrm{TT}} = \frac{P_{\mathrm{GR}}}{P_{\mathrm{PM}}}. \tag{9}$$

3.2 Numerical determination

The ratio is evaluated at a fiducial double neutron-star-like system: $M_1 = 1.4\,M_\odot$, $M_2 = 1.3\,M_\odot$, $a = 10^9\,\mathrm{m}$. Because Equations (7) and (8) share identical functional forms—the PM TT sector was constructed to reproduce GR gravitational-wave predictions—the ratio is

$$k_{\mathrm{TT}} = 1.0. \tag{10}$$

3.3 Physical interpretation

$k_{\mathrm{TT}} = 1$ confirms that the PM quadrupole power formula is numerically identical to the GR Peters formula at the order implemented. This is by design: the TT projector and the 32/5 pre-factor are imported directly from the linearised GR derivation. A value differing from unity would indicate an inconsistency in the PM TT sector that would need to be resolved before the theory could be used for gravitational-wave predictions.

4 Calibration of k_{Fizeau}

4.1 Physical context

When a gravitational lens moves transversely with speed $v \ll c$, the deflection angle acquires a first-order correction relative to the static result. In GR (with the standard PPN parameter $\gamma = 1$) the correction is

$$\alpha_{\mathrm{moving}} \approx \alpha_{\mathrm{static}} \left(1 + \frac{v}{c}\right) + \mathcal{O}\left(\frac{v^2}{c^2}\right). \tag{11}$$

In the PM framework this correction is parametrised as

$$\alpha_{\mathrm{PM,moving}} = \alpha_{\mathrm{static}} \left(1 + k_{\mathrm{Fizeau}} \frac{v}{c}\right), \tag{12}$$

where k_{Fizeau} is the Fizeau drag coupling.

4.2 Determination

Matching (12) to the GR result (11) gives directly

$$k_{\mathrm{Fizeau}} = \frac{\left(\dfrac{\alpha_{\mathrm{moving}}}{\alpha_{\mathrm{static}}} - 1\right)}{\dfrac{v}{c}}. \tag{13}$$

With the GR target ratio $\alpha_{\mathrm{moving}}/\alpha_{\mathrm{static}} = 1 + v/c$, this evaluates to

$$k_{\mathrm{Fizeau}} = \frac{(1 + v/c) - 1}{v/c} = 1.0, \tag{14}$$

independent of the chosen test velocity (fiducial value $v = 30\,\mathrm{km\,s^{-1}}$ is used in the code).

4.3 Physical interpretation

$k_{\mathrm{Fizeau}} = 1$ means the PM medium-drag correction to moving-lens deflection equals the GR first-order v/c correction when $\gamma = 1$. Physically, the Fizeau drag of light by a moving medium is the non-relativistic analogue of the gravitomagnetic contribution to bending in GR. The unit value is consistent with the relativistic aberration / frame-dragging picture and acts as a sanity check that the PM kinematic sector correctly reproduces the PPN $\gamma = 1$ moving-lens result.

Note: the designation "Fizeau" refers to the 19th-century Fizeau drag experiment (partial entrainment of light by a moving medium), and by analogy to the partial frame-dragging of null rays by a moving gravitational lens in the PM index picture.

5 Summary

Constant	**Value**	**Method**	**Refere**
μ_G	$1.4861356 \times 10^{-27}\,\mathrm{m^2\,kg^{-1}}$	Bisection fit: PM path-integral $\to 4GM/c^2b$	Sec. 2
k_{TT}	1.0	Ratio: PM quadrupole power / GR Peters power	Sec. 3
k_{Fizeau}	1.0	Algebraic: PM moving-lens model $\to$ GR v/c correction	Sec. 4

All three constants reduce, up to numerical precision, to values expected from linearised GR with $\gamma = 1$. This provides a self-consistency check: the PM framework recovers standard weak-field GR predictions in the appropriate limits when its index coupling is fixed to $\mu_G = 2G/c^2$.

Code reference: `legacy/Pushing-Medium/tests/fixtures/calibration.py`, function `load_or_calibrate()`.